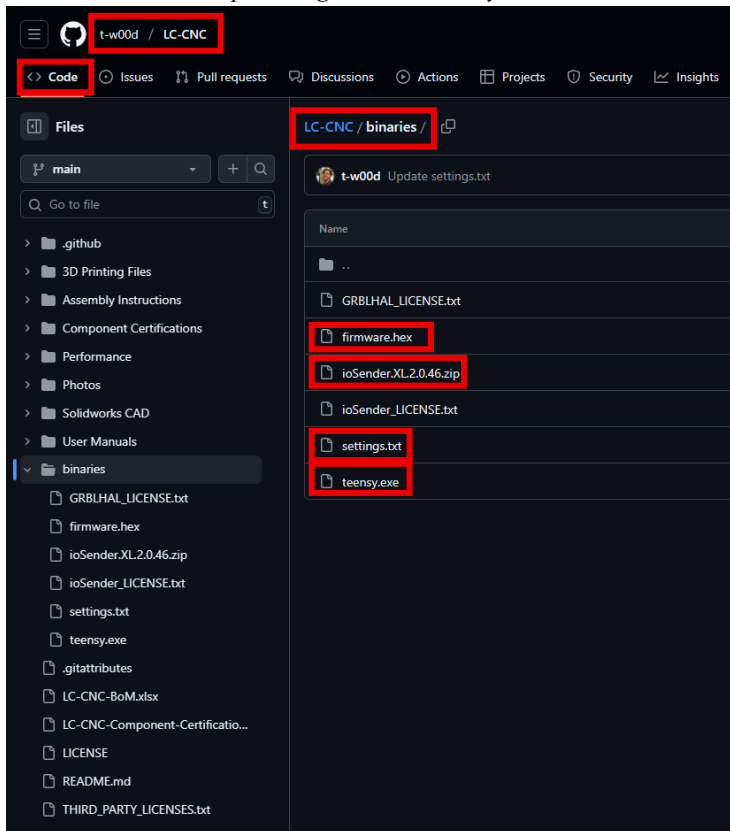


Software instructions

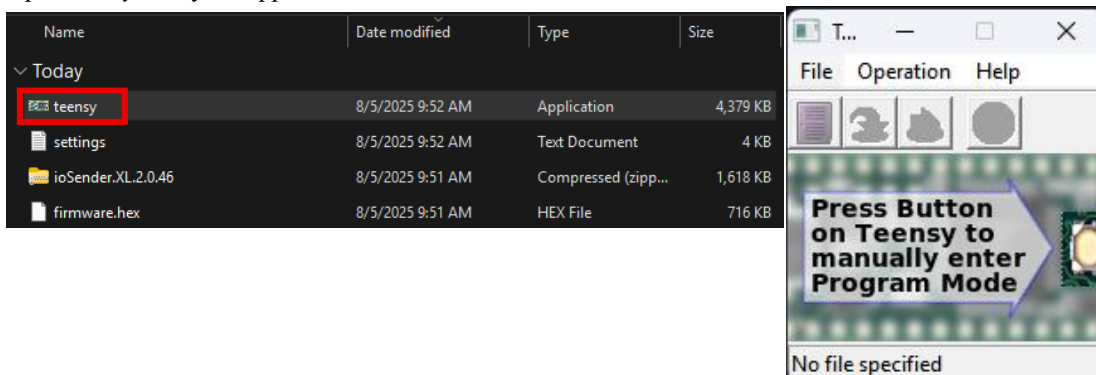
1. Ensure that your computer runs the Windows operating system. In browser, navigate to <https://github.com/t-w00d/LC-CNC>. Under *Code*, open the *binaries* folder and download the files: *firmware.hex*, *ioSender.XL.2.0.46.zip*, *settings.txt*, and *teensy.exe*.



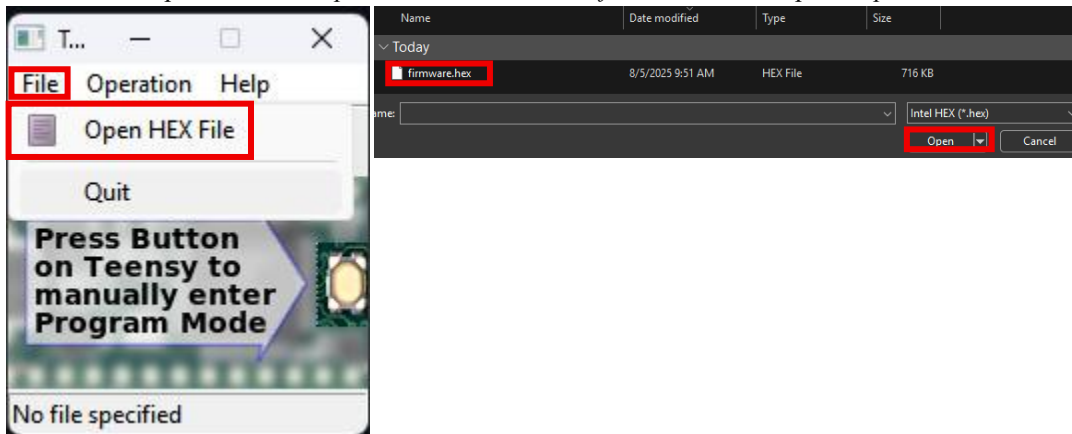
2. Remove teensy from electrical box, plug in usb A end of the cable into your computer and connect the micro usb end to the teensy



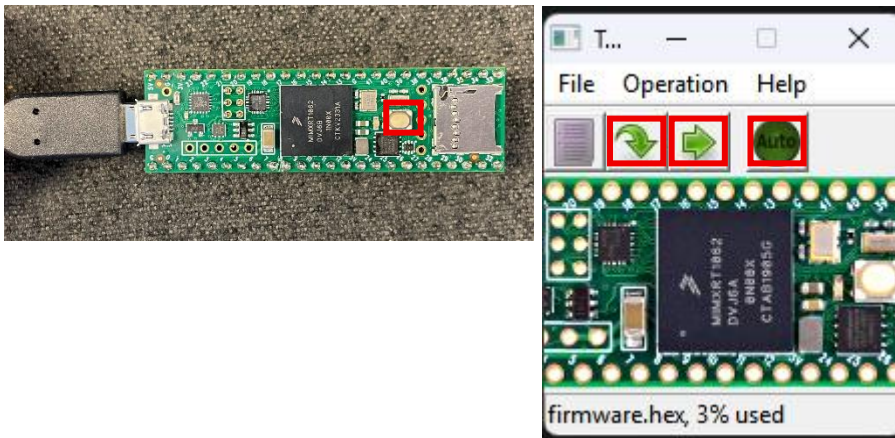
3. Open *teensy.exe*, your app should look as shown:



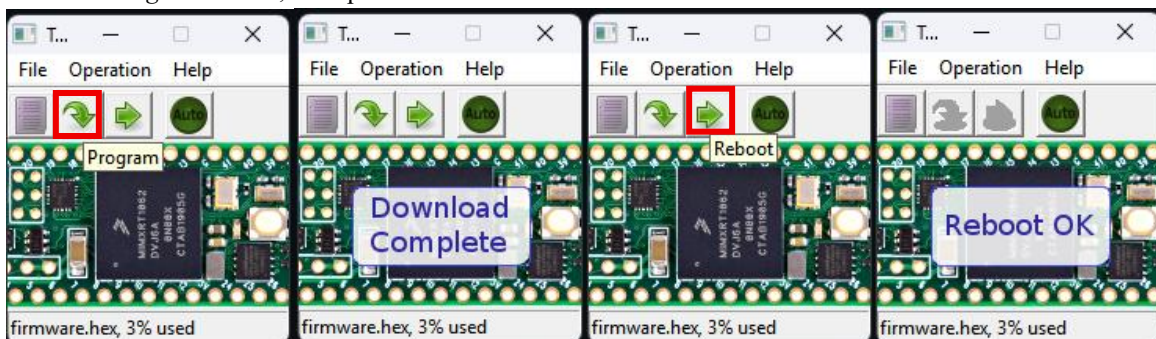
4. Under *File* dropdown, select *Open HEX File* and select *firmware.hex* and press *Open*.



5. Press the button on the Teensy to manually enter program mode, the light should glow red and the software should activate three buttons



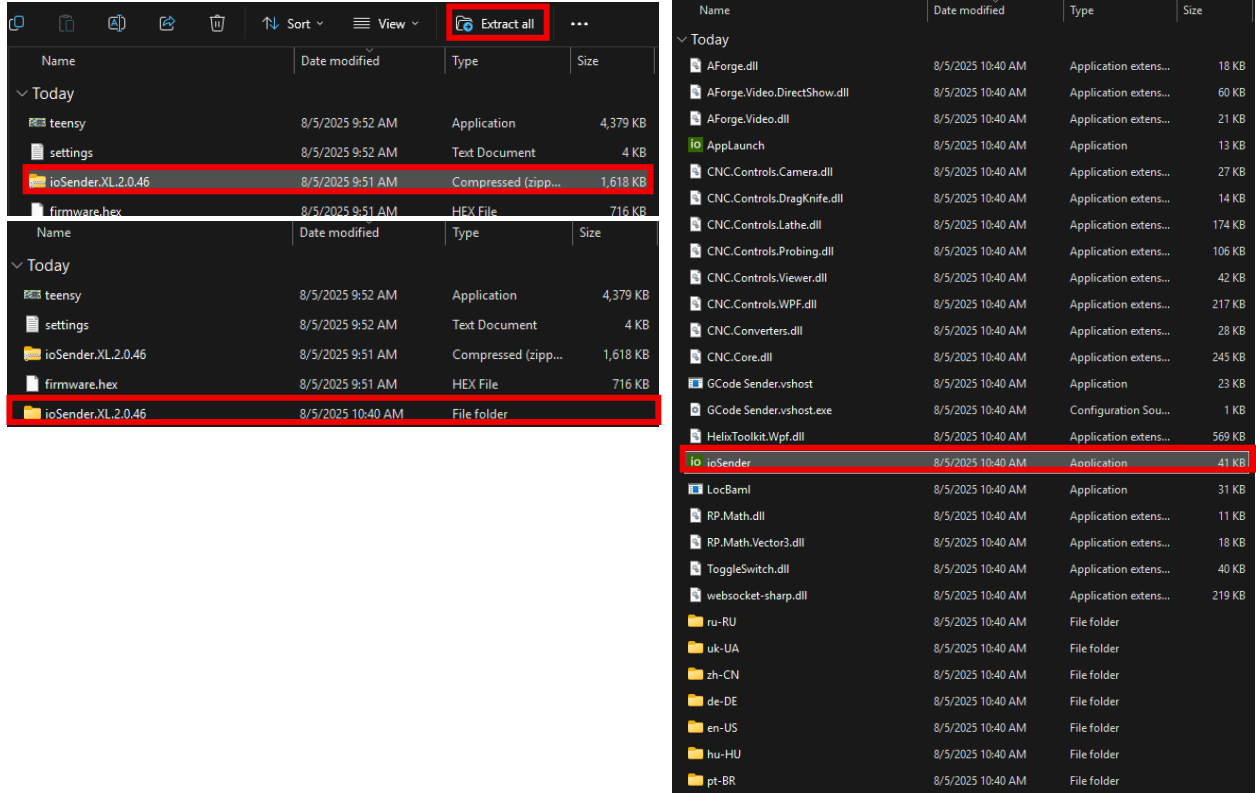
6. Press the *Program* button, after press the *reboot* button.



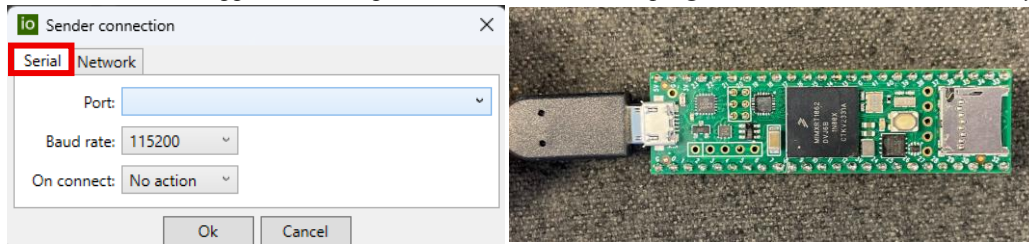
7. Now that the teensy is programmed, unplug the teensy from the micro usb cable.



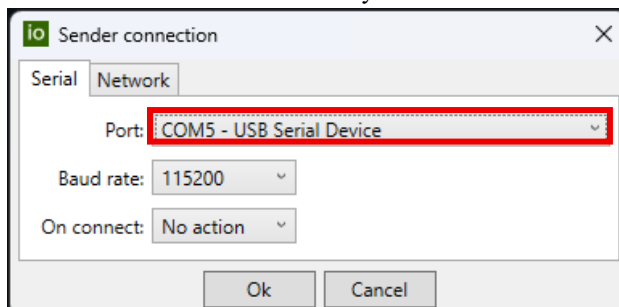
8. Extract the contents from *ioSender.XL.2.0.46.zip*. Once contents are extracted, open the *ioSender.XL.2.0.46* folder and open the *ioSender.exe* application.



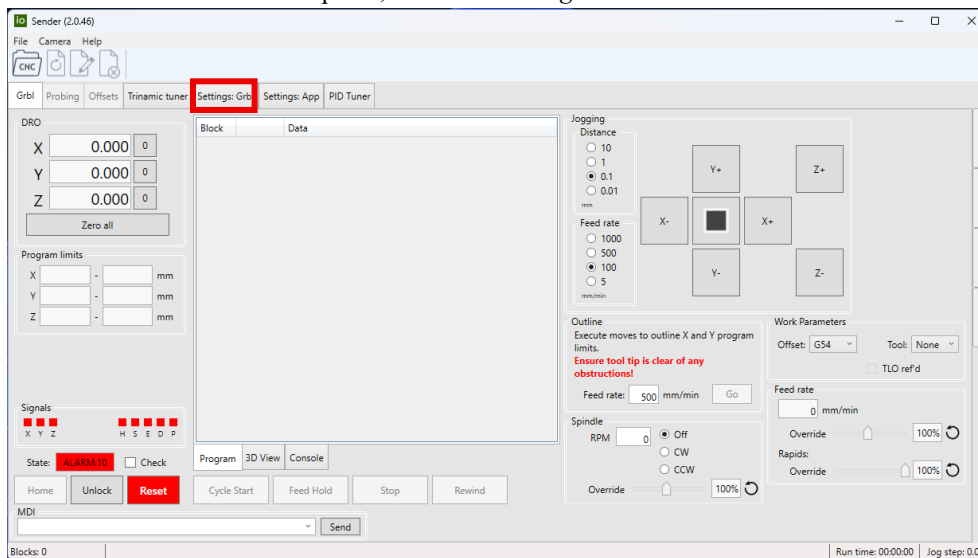
9. Once the *ioSender* application is open and in the *Serial* tab, plug in the micro usb into the teensy.



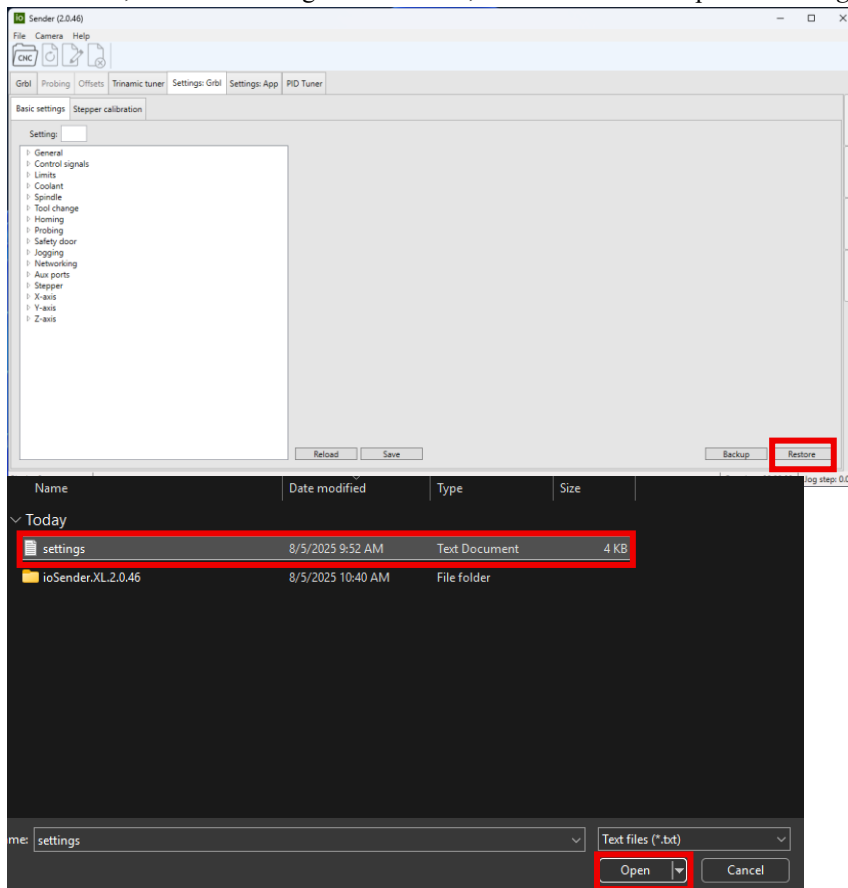
10. Under the *Port* drop down, select the *COM[] – USB Serial Device* and press *Ok*.
Note: The number after *COM* may not be the same as the instructions, this is ok!



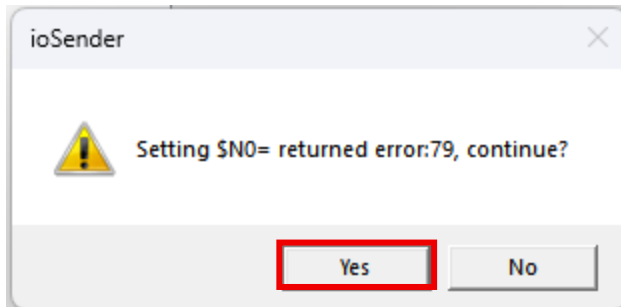
11. Now on the ioSender control panel, select the *Settings: Grbl* tab.



12. On this tab, in the bottom right hand corner, select *Restore* and open the *settings.txt* file.

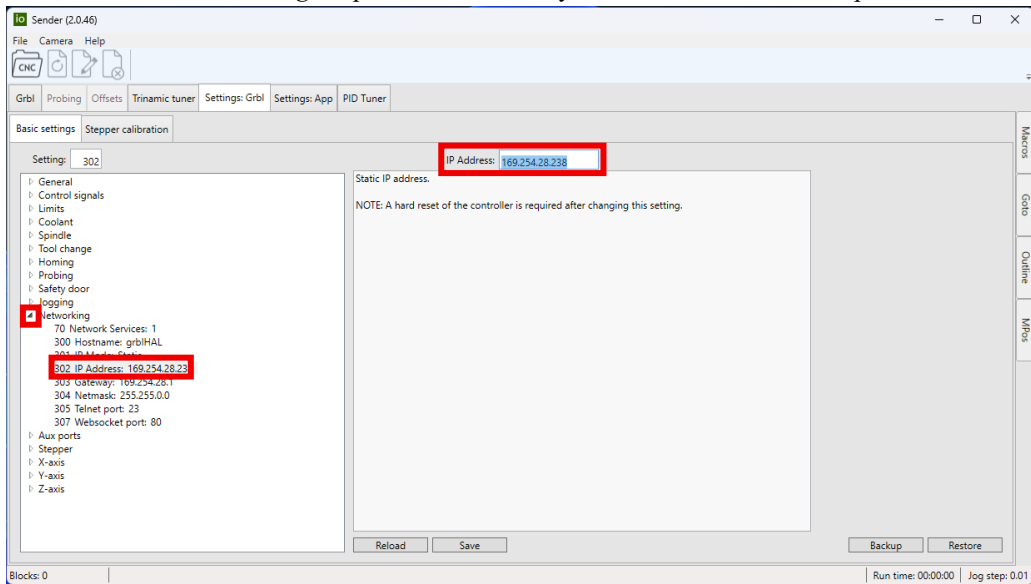


13. If you get this: *Setting \$NO= returned error:79, continue?* Continue by selecting *Yes*. If no error occurs, it will accept it.



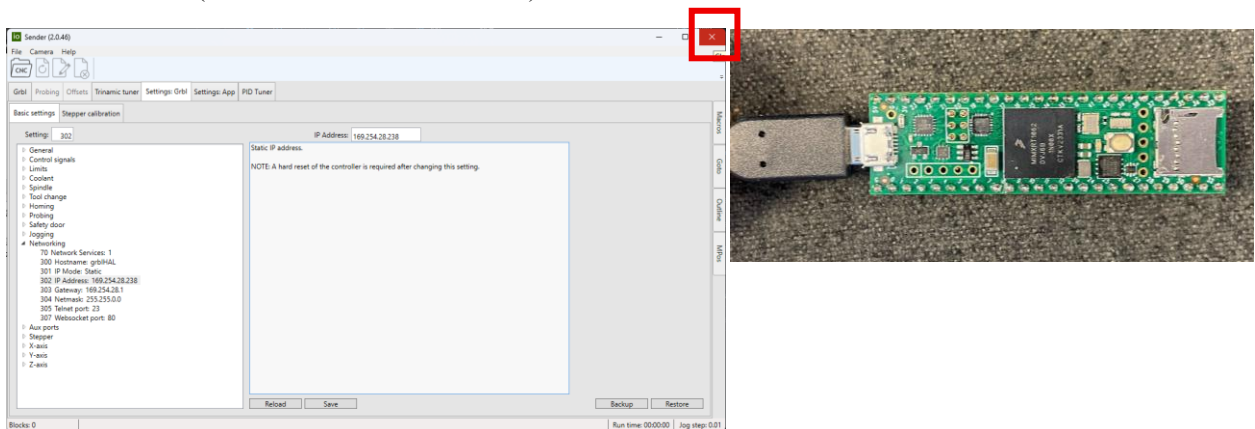
14.

15. Now, extend the *Networking* drop down and record your *IP address* for next steps.

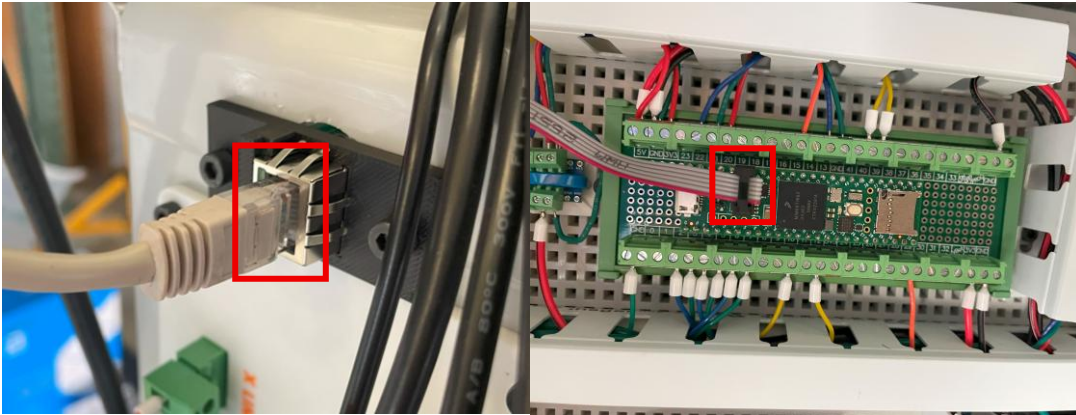


16.

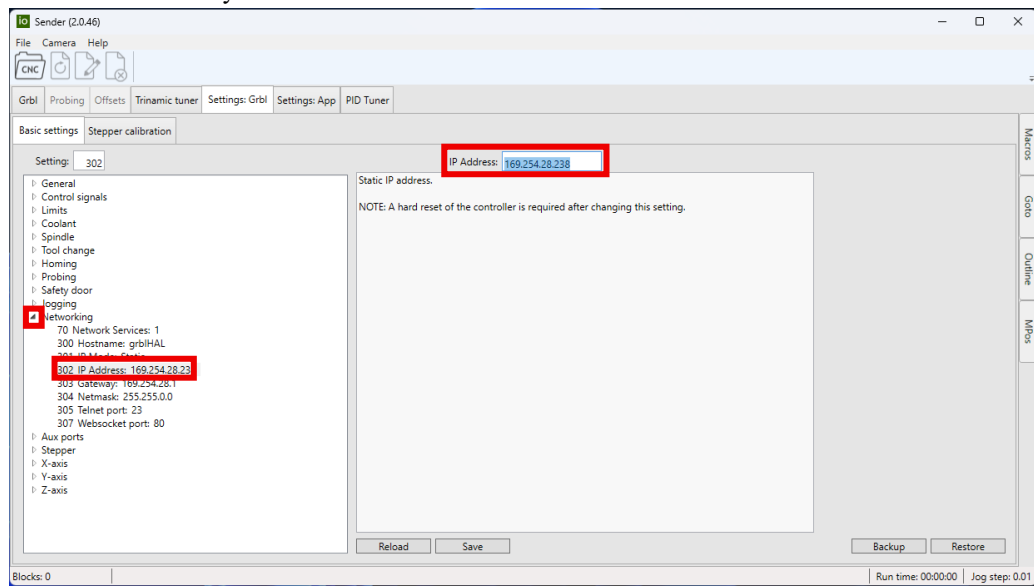
17. Now that you have recorded your IP address, close ioSender, unplug your teensy, then install your teensy into the electrical box (refer to *Electrical Instructions*)



18. Ensure your ethernet cable is plugged into the teensy and your computer.



19. Open ioSender and navigate to *Settings: Grbl* and extend the *Network* drop down and ensure your IP address is the same as the one you recorded earlier.



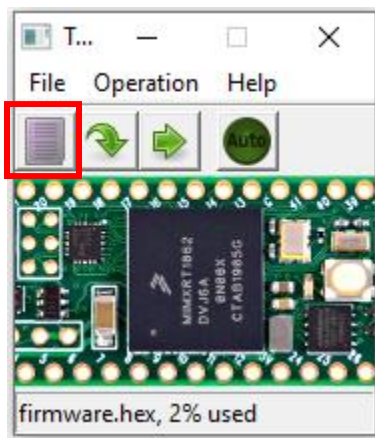
1. Plug in the USB cable from the electrical box to a Windows laptop.



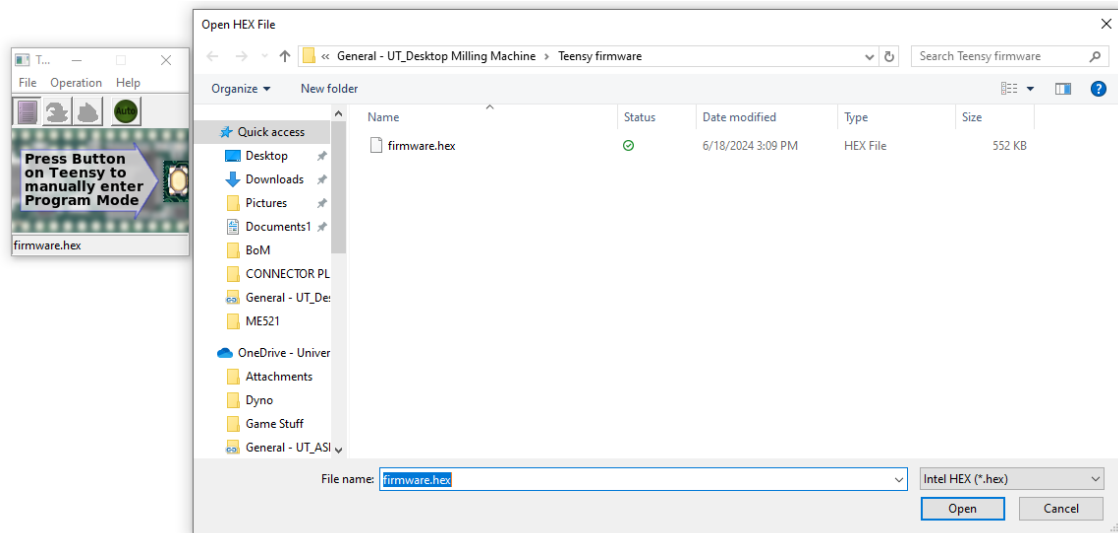
2. Open the file “teensy.exe”.



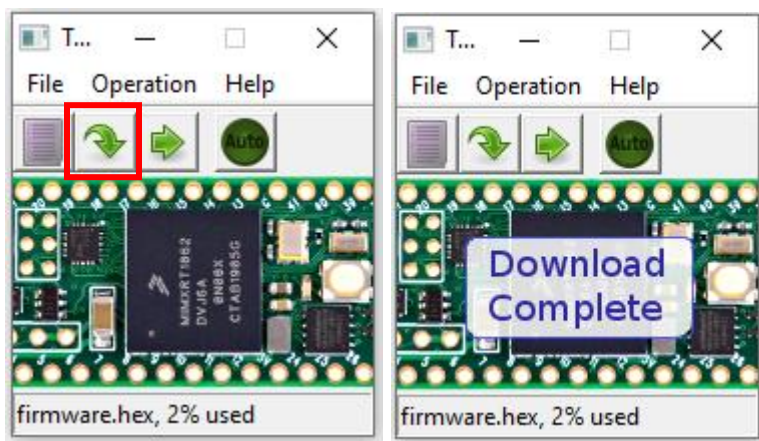
3. Press button on Teensy to manually enter program mode. Select the option outlined in red to select the firmware.



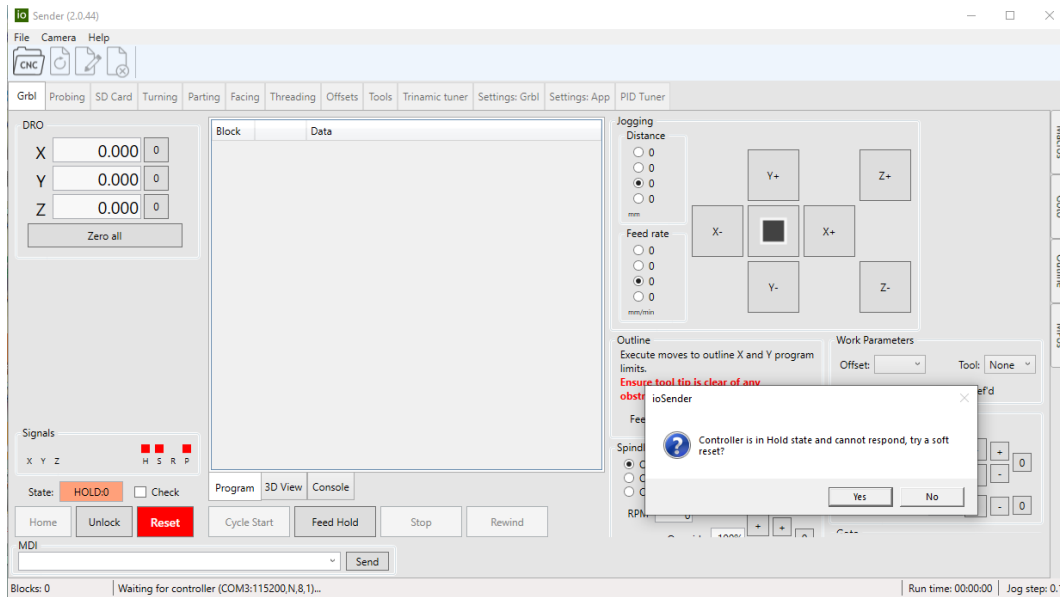
4. Select “firmware.hex” that was provided. Make sure the you select the firmware from the extracted folder and not the zip folder.



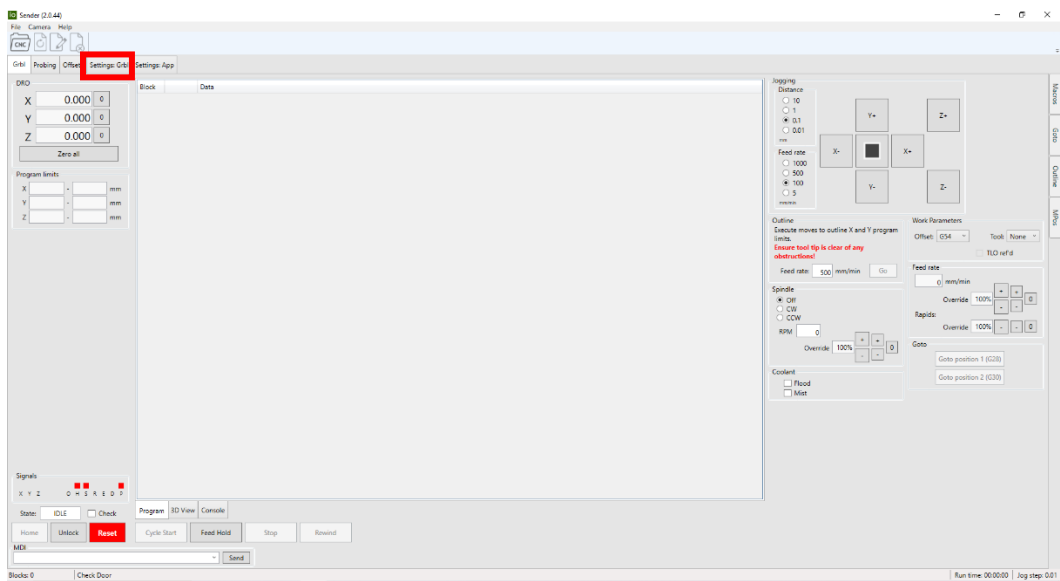
- Once the firmware is selected, select the program button outlined in red.



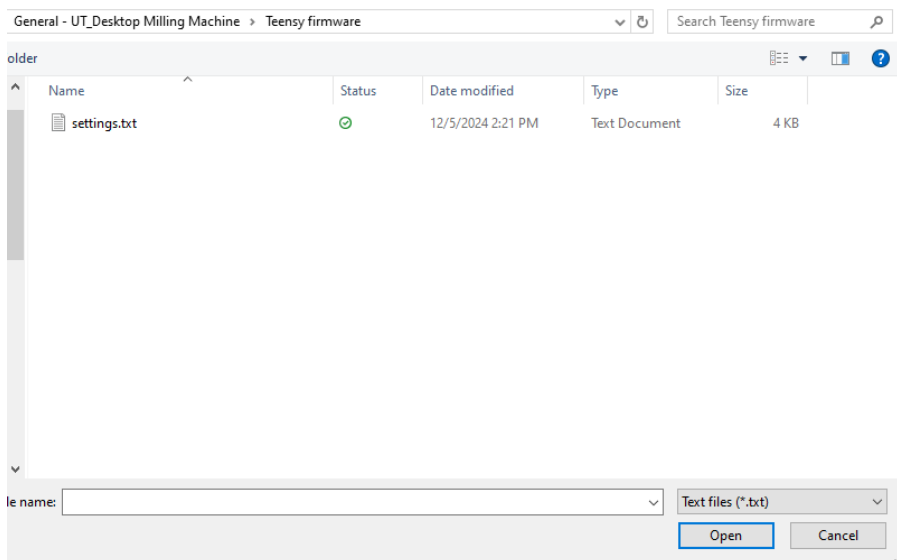
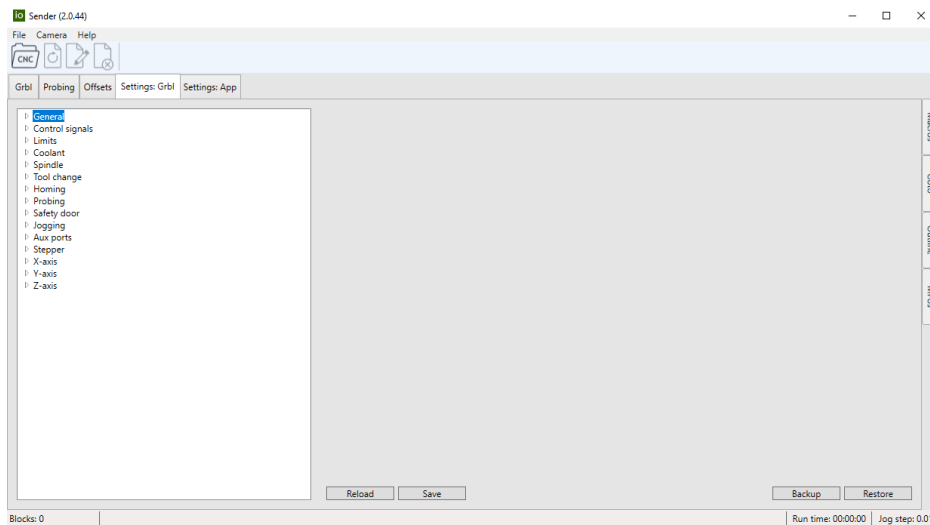
- Download the zip file provided. "ioSender XL 2.0.44.zip" and Extract the folder inside the zip file to the desktop of the laptop.
- Open the extracted folder and open "ioSender.exe". The controller may need a soft reset upon first launch. Select yes if this message is presented.



8. Select “Settings: Grl”



9. Select “Restore” and select the “settings.txt” file provided.



10. The Teensy 4.1 is now set up to run G-code with the proper settings specific to this machine.
11. Next, we will program the VFD with the proper settings for our electrical configuration. Begin by plugging the IEC cable to the VFD port on the small connector plate. Switch on the receptacle.



12. Follow the instructions in this video to learn how to change VFD settings. Do not use the settings provided in the video. After learning how to adjust the settings using the “Set” button, input the following settings.

Note: If pressing the “Set” button doesn’t change the display, press the “Stop” button and try again. Cycle VFD power after settings have been changed.

Video: https://youtu.be/J6nqaAiBo0M?si=FIONAb2tYMn_0Hlu&t=230

F001: 1
F002: 1
F014: 6
F044: 01
F070: 01
F140: 2.2
F142: 8.6
F170: 02
F171: 03